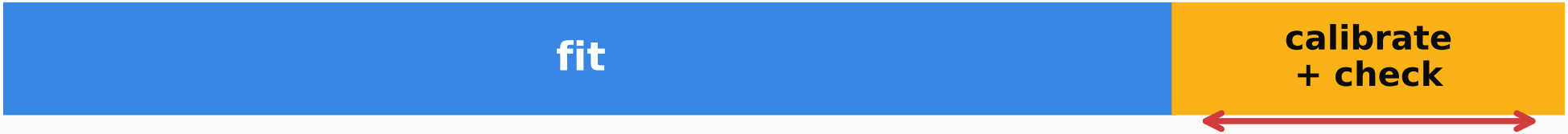
The two-window version, which was wrong:



calibrated AND verified on the same window: the rate
equals the target by construction

The three-window version, split by TIME not at random:

time

a later window the threshold
has never seen

and the gate is a 3-sigma POISSON band on the alarm COUNT, not a multiple of the rate:
on 175 frames at a 1 % target, 1.8 alarms are expected and noise alone spans 0-6.